

Vim in VSCode

An Introduction to Vim Motions

Target Audience: UCF Computer Science Students

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1. Introduction: What Does "Vim" Even Mean?

Vim, which is short for Vi Improved, is a widely used code editor designed to be primarily utilized through the keyboard. The program was written as an evolution of an ancient code editor named Vi. Released in 1991 by *Bram Moolenaar*, Vim is distinct compared to traditional editors due to its philosophy of **Modal Editing**. By utilizing this unique methodology within the familiar environment of VSCode this guide will jump-start your ability to produce code in a manner previously unachievable.

NOTE

While modal editing might initially appear unfamiliar, it is common in numerous varying genres of creative software such as: *Photoshop*, *Adobe Aftereffects*, and *Blender*.

Modal editing is the concept of editing data through utilizing specific tools, or modes. These modes all transform their respective mediums in different manners, allowing users to perform the exact alterations that they desire. A mode can be as simple as the paint brush tool in an image editor or as complex as *Blender's* Transform Operator. In contrast, the traditional text editor only performs one action when a user interacts with its medium of text, that is inserting or removing text.

For this guide we will be installing a plugin to the popular editor VSCode. There are many different ways to access the capabilities of Vim, but for simplicity the vim extension will serve as an excellent starting point to your Vim journey. Instead of entering text into native Vim though, it'll be received by the VSCode environment. This extension will essentially replicate how control & input function within the Vim application, without any of the typical terminal Vim baggage.

Please follow along to the following instructions presented. Vim, like any other tool, requires an initial amount of practice to begin using it in your everyday programming. Thus, you will benefit greatly by performing my instructions as I introduce the application. Once you do get the hang of it, you'll see just how incredibly efficient one can get through practice and determination.

2. Installation & Setup in Visual Studio Code

Step 1: Open the Extensions Marketplace

To begin, we must download the Vim extension from the marketplace. In the left sidebar, click on the lowest icon. It will look like four squares in a grid, with the top right corner rotated slightly. This is where you can install numerous extensions that enhance the capabilities of VSCode.

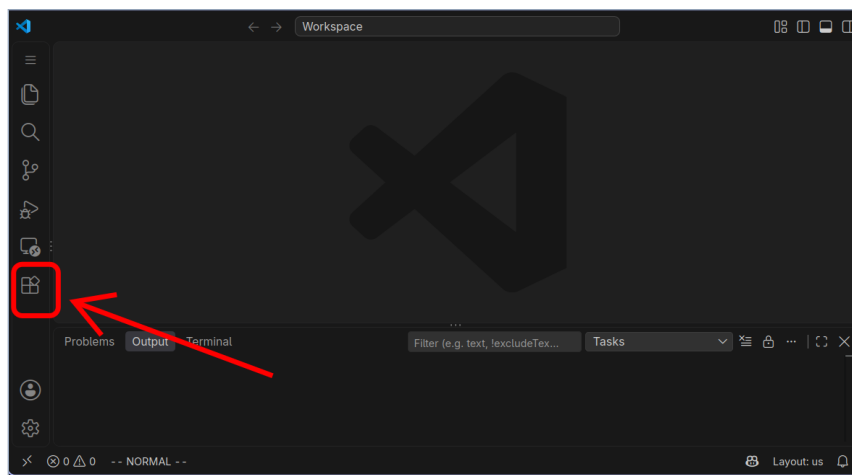
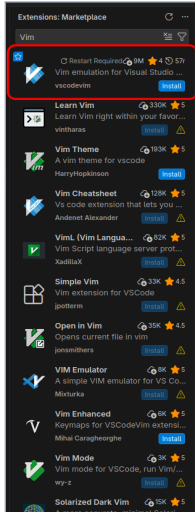


Fig. 1. Visual Studio Code Activity Bar indicating the Extensions Marketplace icon.

Step 2: Search and Install VSCoDeVim



Once you have opened the menu, you will see a search bar on the top of the panel. Click on the search bar, type in "Vim", and hit 'enter'. Once the results have loaded you'll want to find the extension labeled 'Vim', with the subtext of 'Vim emulation for Visual Studio Code'. It will have a blue background and a large white 'V' as the logo. Then click the blue install button that is on the bottom right of the extension. There will be many different options that look similar at first glance, but do not worry about those. Just install the extension I have shown in this guide and you will have all the capabilities of Vim at your fingertips.

WARNING!

Once installed, VSCoDeVim activates immediately without requiring an editor restart. Your editor will now behave like Vim, so be prepared to be unable to edit code in the traditional manner.

Fig. 2. Extensions Marketplace search results highlighting the official VSCoDeVim package.

3. The Modal Architecture

A modal architecture requires the user to swap through many different modes, each with differing behavior, throughout a session. Each mode is designed to perform a specific type of action, like mass editing of a large sections document or precise character input in a single sentence. In a image editing program, the user switches modes and selects tools by clicking on buttons with a mouse. While intuitive and easy to learn for newcomers, this methodology does not properly utilize the efficiency presented by the keyboard. The mouse primarily has three buttons for a user to operate their program, while the keyboard can have up to 104! Each button can be pressed in a near equal amount of time and effort from the user. This allows for an enormous range of actions to be performed on each key press. Using this philosophy a power user can manipulate his document in the exact manner in which he desires, in a rapid fashion, all without ever having to slow down by reaching for the mouse.

To switch modes in Vim, you must first be in the 'NORMAL' mode, and then hit the specific key tied to your desired mode. To return to NORMAL mode either hit the 'Escape' key or Ctrl-C. Each key performs a different action depending on the user's selected mode. The mode is displayed in a status bar at the bottom of the screen.

Core Editing Modes

- **Normal Mode:** The default state of Vim. All keypresses are interpreted as navigation or manipulation commands rather than literal characters.
- **Insert Mode:** The traditional editing mode where keys that are typed are placed on the screen, as compared to controlling the program.
- **Command-Line Mode:** A special mode that allows users to enter powerful commands. Entered by pressing `:` from NORMAL mode.
- **Visual Mode:** Used for highlighting text over multiple lines, it replaces selecting text by dragging the mouse. Entered by hitting `v` which highlights individual characters or `V` which highlights an entire line.

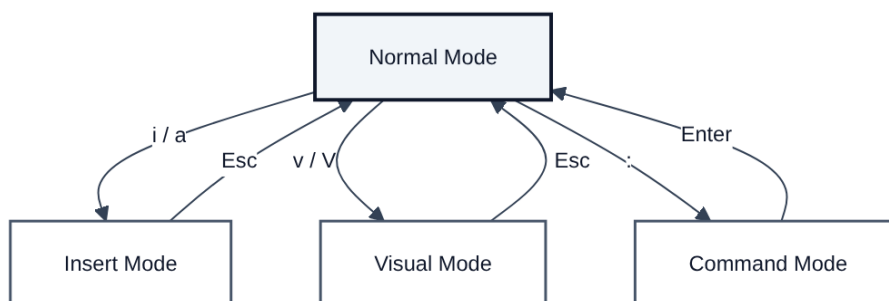


Fig. 3. Vim modal architecture state transitions and keybindings.

4. Navigation

In order to navigate without utilizing the mouse, you must use the keyboard in normal mode to move the cursor around. Each key in normal mode has a function, but you do not need to know all of them. We're going to cover the keys fundamental to basic movement through your document. It is essential that you place your hands in the *Touch Typing* position, with each index finger on 'f' and 'j'.

Keystroke	Movement Direction	Functional Description
h	Left	Moves cursor one character left.
j	Down	Moves cursor one line down.
k	Up	Moves cursor one line up.
l	Right	Moves cursor one character right.
w	Word Forward	Advances cursor to the beginning of the next word.
b	Word Backward	Moves cursor backward to the start of the previous word.
f	Find	Jumps directly to the character specified after. Holding shift jumps in the reverse direction.
0	Line Start	Jumps directly to the first character of current line.
\$	Line End	Jumps directly to the last character of current line.
gg or go	File Top	Moves cursor immediately to the first line of file.
G	File Bottom	Moves cursor immediately to the last line of file.

GO PRACTICE!

Take the time now to practice your navigation in a document. There is a learning curve but once you performed the movements a couple times the skills will develop and it will feel completely natural.

5. Editing

Now that you've learnt to navigate a document we have to start editing it. There are a couple of important editing commands in normal mode, but not nearly as many as those required for moving the cursor. Importantly, hitting 'i' will place you into insert mode and 'a' will place you into insert mode, but after your current cursor. If you hold shift while using those keys your cursor will jump to the beginning or end of the line respectively.

Keystroke	Operator Action	Operational Scope
x	Delete Character	Deletes single character directly under cursor.
dd	Delete Line	Cuts current line into default clipboard register.
yy	Yank (Copy) Line	Copies entire current line without modifying text.
p	Put (Paste)	Pastes copied/deleted text immediately after cursor.
u	Undo	Reverses the most recent editing action.
<Ctrl-r>	Redo	Reapplies changes previously reversed by undo.
r	Replace Character	Replaces single character under cursor with next keypress.

NOTE

The operator 'u' will be your *best friend*. If you ever accidentally hit a key, or made a mistake just hit 'u' as fast as possible. It is common to make many mistakes while learning, so do not be afraid to utilize undo as often as possible.

6. Operator Combination

Now we have covered the skills required to edit a document, everything presented in the following sections is designed to increase your efficiency exponentially. Now I have not explained the last core component of the Vim philosophy, **you can chain operators together**. Essentially almost every operator is designed to be paired with another of the same type, ie: navigation, editing, or control. This requires a couple of examples to fully grasp its power. The simplest example of this behavior is by utilizing number with movement keys.

In NORMAL mode, hit a number and then hit a movement key (h, j, k, l). Your cursor will be moved as many times as you chose the number! This also applies for any other command. You could even paste 100 times in one button press if you wanted to!

The real efficiency boon comes from chaining selection commands (w, b, \$, or t) and editing commands like d or c.

NOTE

The 'c' key, which represents the *change* action, will remove a selected piece of text and place the user in insert mode. Essentially allowing them to change whole words or sentences in just two keypresses.

Chain	Operator Action	Operational Scope
dw	Delete Word	Deletes an entire word, starting from the cursor.
cw	Change Word	Deletes a word and places the user in INSERT mode.
5i	5x Insert	Will paste whatever text is typed in insert mode 5 times.
db	Delete Back	Deletes behind the cursor to the start of a word.
d\$	Delete To End Of Line	Deletes to the end of a line.
ct{key}	Change to Character	Deletes up to a character, then places the user in INSERT mode. Using this to a period allows for lightning fast editing of a sentence.
ci{key}	Change in Character	Deletes all content surrounded by character, then places the user in INSERT mode.

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Rhetorical Rationale

The primary goal of my document is to serve as an introduction to the code editor *Vim*, which is a niche tool designed to edit source code in a manner that minimizes the usage of the mouse. This tool requires a dense set of knowledge for even novice usage, which serves as a barrier to adoption for the general programming population. Thus it is my goal to author a professional document in order to assist my fellow programmers in learning to use the program, which should enhance their efficiency and grow the local community surrounding a personal interest.

The genre of *Technical Guide* was ideal for conveying the skills required to begin using *Vim*. These types of documents compact a large amount of information into a functional format that provides their audiences with information that they can immediately use. The topic chosen is primarily a functional one, and thus requires a functional manner of document to properly convey its intricacies. When learning a new skill one must practice, growing their ability in small portions until they eventually master it. A guide presents information in a serialized format, which builds an understanding of the program incrementally. This gradual growth is essential to beginning to use *Vim*, as attempting to learn the entire program initially is impractical.

The audience that my *Professional Information Project* is designed for are my peers within the School of Computer Science at UCF. Specifically those who have prior experience utilizing code editors like *VS Code* or *CLion* to complete their school assignments. This group originates from a technical background and is comfortable interacting with their computer through advanced methods.

The content presented within is presented as a means in which to increase one's efficiency in writing code. The overarching promise of performance enhancement will serve as the main draw behind the document and the practice required to properly utilize the advice provided. While the surface level use of *Vim* can be challenging to learn in a single sitting, overall the amount of knowledge required to use it is manageable. This amateur use of the application is the goal of the document, but the tool itself contains leagues of depth and complex behavior. Much like other examples of professional software, for instance *Photoshop* or *Blender*, it would take thousands of hours of investment to master *Vim* in its entirety. Such an environment is bound to draw inquiry from an audience that is engaged with the content, as each presentation of introductory features can easily lead to a multitude of supplementary questions.

The guide prioritized basic usage